



IPM Package of Practices for Banana

(For producing quality fruits for export)



Government of India

Ministry of Agriculture & Farmer's Welfare

Department of Agriculture & Farmer's Welfare

Directorate of Plant Protection Quarantine & Storage

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IPM Package of Practices for Banana (For producing quality fruits for export) was compiled by the DPPQS technical team under the Chairmanship of **Dr. Ravi Prakash, Plant Protection Adviser, and guidance of Dr. Pramod Kumar Meherda, IAS, JS (PP). Dr. S.C. Dubey, ADG(PP), ICAR, New Delhi** provided review from crop specific ICAR institute - National Research Centre for Banana, Tiruchirapalli.

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FOREWORD

Pests and disease menace cause significant crop losses to the farmers. Generally, farmers apply large quantity of pesticides indiscriminately to control pests & diseases during various crop growth stages either calendar based or on noticing damage symptoms in the crop. Indiscriminate uses of chemical pesticides are a major cause of ecological imbalances, environmental pollution, pesticide resistance, pest resurgence and pesticide residues.

Integrated Pest Management (IPM) is a globally accepted strategy for promoting sustainable agriculture. Initially IPM strategy allow chemical pesticide application on pest population reaching at economic threshold level which has shifted to more ecologically sustainable Agro-Eco System Analysis (AESA) based decision making for selection of IPM technique. The AESA based decision making consider relationship among various components of an agro-ecosystem with special focus on pest-defender dynamics, abilities of plant to compensate for the damages caused by the pests and the influence of abiotic factors on pest buildup. IPM approach advocate utilization of alternate pest management techniques like cultural, mechanical and biological prior to use of chemical pesticides. Pesticide residue apart from pest association in Agriculture produce has proved major impediment for export especially of fresh fruits and vegetables.

Banana is one of the important export potential fruit of India grown in very large area. Sincere efforts have been made by DPPQS & ICAR to incorporate all the aspect that will help in production of pest as well as pesticide residue free Banana that not only help in promoting export but also provide healthy fruits to Indian citizen. I hope that IPM Package of Practices for Banana (For producing quality fruits for export) will be a handy document to Central and State government functionaries involved in extension and farmers for production of residue free quality Banana fruits.



Dr. Pramod Kumar Meherda
Joint Secretary (Plant Protection)

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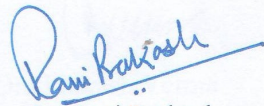
FOREWORD

IPM is a science-based, decision-making process that combines biological, cultural, physical and chemical tools to reduce damage from pests and pest management tactics in a way that minimizes overall economic, health and environmental risks.

During the late 90s FAO started advocating Agro-Ecosystem Analysis (AESA) based IPM. Experience in different countries have since show that AESA, which takes into account ecological principles and relies on the balance that is maintained by biotic factors in an ecosystem has also resulted in reduction in cost of production and increase in yields. AESA based IPM also takes into account the need for active participation of farmers and promotes experiential learning and discovery based decision making by farmers. AESA based IPM in conjunction with ecological engineering for pest management promotes bio-intensive strategies as against current chemical intensive approaches, while retaining the option to apply chemical pesticides judiciously as a measure of last resort.

The resource persons of DPPQS have made sincere efforts in IPM Package of Practices for Banana (For producing quality fruits for export) by incorporating agro-ecosystem analysis, ecological engineering, pesticide application techniques, other IPM options and post-harvest management for the pesticide residues free in banana and to export of banana with the crop based plant protection measures. I convey my sincere thanks to Dr. S.C. Dubey, ADG(PP), ICAR, New Delhi for providing review from crop specific ICAR institute - National Research Centre for Banana, Tiruchirapalli.

I hope this IPM package will serve as a ready reference for extension functionaries of Central / State Governments, NGOs and farmers for producing exportable banana fruits.



Dr. Ravi Prakash
Plant Protection Adviser

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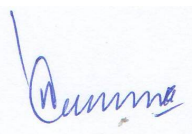
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PREFACE

India is the largest producer of banana in the world with the production of 30.5 million tonnes and share 23.7% of total world production. However, India has negligible share on Banana fruit export. Phytosanitary and Sanitary compliance are major aspects for export of any fresh fruits. Indiscriminate use of chemical pesticides for management of pests/ diseases results in pesticides residue in fruits and fails meeting sanitary requirement of various importing countries. In recent time, most of the countries incorporating Maximum Residue Level (MRL) as major sanitary requirement for import of Agricultural commodities especially for fresh fruits & vegetables. So there is challenge to produce pest as well as pesticide free fruits for export. Adoption of Agro-ecosystem Analysis (AESA) based Integrated Pest Management (IPM) can promise production of pests as well as pesticide residue free fruits and comply with Phytosanitary and Sanitary requirements of most of the Banana importing countries. In addition, AESA based IPM also conserves bio-diversity in agro-ecosystem, reduce environmental pollution and reduce cost of cultivation and thereby increase farmers' income.

I convey my sincere thanks to Joint Secretary (Plant Protection) & Plant Protection Adviser for encouragement and facilities for compilation of this PoP and also to ADG (PP), ICAR, New Delhi for review from crop specific ICAR institute - National Research Centre for Banana, Tiruchirapalli.

This IPM Package of Practices for Banana (For producing quality fruits for export) will prove an important guide for extension functionaries of Central / State Governments, NGOs and farmers for producing exportable banana fruits.



Om Prakash Verma
Joint Director (IPM)

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1. INTRODUCTION

Banana (*Musa* spp.), basically a tropical crop, grows well in a temperature range of 15°C-35°C with relative humidity of 75-85%. Presently it is being grown in more than 140 countries, mainly in tropical and subtropical regions. The fruit is recognized as the fourth most important global food commodity. India's share is 32 per cent of the total fruit production.

Processed products, such as chips, banana puree, jam, jelly, juice, wine and halwa can be made from the fruit. The tender stem, which bears the mixed spadix or Thyrsus type of inflorescence is extracted by removing the leaf sheaths of the harvested pseudostem and used as vegetable. Plantains or cooking bananas are rich in starch and have a chemical composition similar to that of potato. Banana fiber is used to make items like bags, pots and wall hangings. Rope and good quality paper can be prepared from banana waste. Banana leaves are used as healthy and hygienic plates for serving foods and packing of food stuffs.

Currently, India is the largest producer of banana in the world with the production of 30.5 (world production: 125.3) million metric tons from 8.6 lakh ha area (world area: 60.2) and 34.0 metric tonnes/ha productivity (world productivity: 19.2). The top banana-producing countries of the world are India (23.7% of total production), China (8.9%), Indonesia (6.3%), Brazil (5.4%), Ecuador (5.3%), Philippines (4.7%), Angola (3.1%), Guatemala (3.1%) and Colombia (3.0%).

2. CLIMATIC AND SOIL REQUIREMENTS

Banana prefers tropical humid low lands and is grown from the sea level to an elevation of 2000 m above mean sea level. Chilling injury occurs at temperature below 12°C. High velocity of wind which exceeds 80 km /hr severely damages the crop. Four months of monsoon (June to September) with an average 650-750 mm rainfall are most important for vigorous vegetative growth of banana. Deep, rich loamy soil with pH between 6.5 – 7.5 is most preferred for banana cultivation. Soil for banana should have good drainage, adequate fertility and moisture.

The optimum temperature for leaf emergence is about 25-30°C, the upper and lower limits being about 40°C and 10°C respectively. For optimum growth, the area should have evenly distributed monthly rainfall of 20-22 mm and areas with distinct

long dry season could be developed into good banana producing ones provided sufficient irrigation facilities are available. The crop comes up well from sea level up to an altitude of 1500 m in the tropics. At higher altitudes, banana cultivation is restricted to a few varieties like ‘Hill banana/Virupakshi/Sirumalai’.

3. CULTIVARS AND VARIETIES

Cavendish cultivar like Grand Naine and Robusta are generally preferred for export. However, there is a scope for export of indigenous varieties viz., Yelakki (Ney Poovan), Poovan, Nendran, Red Banana, Karpuravalli, Rasthali, Matti, Sannachenkadali etc.,

Important Banana Varieties Cultivated in Different States of India:

State	Varieties Grown
Andhra Pradesh	Dwarf Cavendish, Robusta, Rasthali, Amritpani, Thellachakkerakeli, Karpooora Poovan, Chakrakeli, Monthan and Yenugu Bontha
Assam	Jahaji (Dwarf Cavendish), Chini Champa, Malbhog, Borjahaji (Robusta), Honda, Manjahaji, Chinia (Manohar), Kanchkol, Bhimkol, Jatikol, Digjowa, Kulpait, Bharat Moni
Bihar	Dwarf Cavendish, Alpon, Chinia , Chini Champa, Malbhig, Muthia, Kothia , Gauria
Gujarat	Dwarf Cavendish, Lacatan, Harichal (Lokhandi), Gandevi Selection, Basrai, Robusta, G-9, Harichal, Shrimati
Jharkhand	Basrai, Singapuri
Karnataka	Dwarf Cavendish, Robusta, Rasthali, Poovan, Monthan, Elakkibale
Kerala	Nendran (Plantain), Palayankodan (Poovan), Rasthali, Monthan, Red Banana, Robusta
Madhya Pradesh	Basrai
Maharashtra	Dwarf Cavendish, Basrai, Robusta, Lal Velchi, Safed Velchi, Rajeli Nendran, Grand Naine, Shreemanti, Red Banana
Orissa	Dwarf Cavendish, Robusta, Champa, Patkapura (Rasthali)
Tamil Nadu	Virupakshi, Robusta, Red Banana, Poovan, Rasthali, Nendran, Ney Poovan, Monthan, Karpuravalli, Sakkai, Peyan, Matti
West Bengal	Champa, Mortaman, Dwarf Cavendish, Giant Governor, Kanthali, Singapuri

Export varieties:

Grand Naine, Robusta, Rasthali, Yelakki (Ney Poovan), Poovan, Nendran, Red Banana, Karpuravalli, Matti, Sannachenkadali

4. LAND PREPARATION

Prior to planting banana, green manure crops like sunnhemp, daincha, cowpea etc. may be grown. The land should be thoroughly ploughed for 2-4 times and leveled. Rotavator or harrow is used to break the clod and bring the soil to a fine tilt. During soil preparation basal dose of FYM (about 50 tons/ha before last harrowing) is added and thoroughly mixed into the soil.

5. PLANTING

Normally side suckers with a well-developed broad base and a pointed tip with narrow, sword-shaped leaf blades at the early stages of the growth and at later stages the leaves emerge normally with broad blades. In general sword suckers emerge close to the healthy mother corms and are more vigorous, grows rapidly and capable of early shooting of bunches and fruit maturity. Hence, sword suckers are preferred as the best planting material. The roots and base of the planting material may be removed. The suckers are dipped in a solution of Carbofuron 3% CG before planting.

5.1 Planting Season:

Planting of tissue culture banana can be done throughout the year as per the market demand except when the temperature is too low or too high. The important seasons for planting followed in different states of India are:

State	Planting time
Maharashtra	· Kharif - June – July
	· Rabi - October – November
Tamil Nadu	· February – April
	· November - December
Kerala	· Rainfed- April-May
	· Irrigated crop- August- September

5.2 Spacing:

Season	Spacing
Kharif	1.5 x 1.5 m., 2 x 2 m. or 2.5 x 2.5 m.
Rabi	1.5 x 1.2 m., 1.5 x 1.37 m.

5.3 High Density Planting:

High density planting is in practice to accommodate 4444 to 5555 plants per ha. Planting 3 suckers / pit at a spacing of 1.8 x 3.6 m (4,600 plants per ha) for Cavendish bananas and 2 x 3 m for Nendran banana (5000 plants per ha) is followed.

5.4 Planting Method:

A pit size of 0.5 x 0.5 x 0.5m is normally required. Small pits are dug in case of ridges and furrows. The pits are to be refilled with topsoil mixed with 10 kg of FYM (well decomposed), 250gm of neem cake and 20gm of carbofuran. Prepared pits are left open for 15-20 days for solar radiation to kill all the insects, soil borne diseases and for aeration before refilling. In saline alkali soil where pH is above 8, pit mixture is to be modified incorporating organic matter and gypsum.

The suckers are planted in the center of the pit and soil around is compacted. Plants are planted in the pits keeping pseudostem 2cm below the ground level. Deep planting should be avoided. The field is irrigated immediately after planting. Furrow planting is practiced in annual planting system in the states of Gujarat and Maharashtra. Trench planting is practiced in wet land cultivation of Cauvery delta region of Tamil Nadu. The pits are filled with top soil, 10-15 kg FYM and 250g super phosphate, mixed well and the pits are closed to about a couple of inches below the ground level.

6. NUTRIENT MANAGEMENT

Apply 25 % N in organic form, 75 % N in inorganic form along with growing and mulching of green manure crops.

- Apply 25% N as farmyard manure and 1 kg neem cake.
- Apply 100g-200g of N/plant as top dressing in three equal split doses 60, 90 and 120 days after planting.
- In acidic soils, triple superphosphate or diammonium phosphate (DAP) is recommended.
- Application of 300-400g K₂O/plant in 3-5 splits is recommended depending upon the cultivar.

Micronutrients

- Combined foliar application of ZnSO₄ (0.5%), FeSO₄ (0.2%), CuSO₄ (0.2%) and H₃BO₃ (0.1%) applied at 3, 5 and 7 months after planting helps to increase yield and quality of banana. Foliar spraying of ICAR-NRCB released banana micronutrient mixture @ 10g/litre of water during 4th, 5th, 6th and 7th month after planting along with good quality surfactant is recommended.

Daily fertigation schedule for tissue culture banana (1200 plants/ac)

Days after planting	Urea Daily (Kg)	Super phosphate	MOP (KCL) Daily (Kg)	Water requirement (lit/plant/day)
30-75 (45 days)	0.60	250.00	0.90	4
75-120 (45 Days)	1.20	-	1.50	8
120-165 (45 days)	2.40	-	3.60	12
165-210 (45 days)	3.00	-	4.80	16
210-255 (45 days)	2.70	-	3.60	20
255-270 (15 Days)	2.40	-	3.00	24

Super phosphate @ 250 gm/plant is applied in the soil as basal dose.

Weekly fertigation schedule for banana (gram/plant/week)

Weeks after planting (Total number of weeks)	Urea	MOP
9 th to 18 th week (10 Weeks)	15	8.0
19 th to 30 th week (12 weeks)	10	10
31 st to 40 th week (10 Weeks)	7.0	12
41 st to 46 th week (5 Weeks)	Nil	10

75% of the recommended dose of Fertilizers (RDF), thus saving 25 % of RDF.

6.1. Nutritional deficiencies/disorders

S. No	Name of the Nutrient	Deficiency/ Disorders	Control	Symptoms
1.	Nitrogen	Leaves become pale green. Mid ribs, petioles and leaf sheaths turn reddish pink and rosette in appearance. Bunch weight and fruit quality is affected.	Application of urea (100g/ plant) followed by irrigation	

2.	Phosphorus	Plants show stunted growth with poor root development. Old leaves show saw tooth marginal chlorosis, curling of leaves, breaking of petioles and bluish green colour of younger leaves.	Application of DAP (50g/plant) followed by irrigation	
3.	Potassium	Orange yellow colour of old leaves, scorching along the margins, reduction in total leaf area, curving of midribs etc. Choking of leaves delay flower initiation	Soil application of 100-150g MOP or foliar spraying of 1.0-1.5 % Potassium sulphate solution	
4.	Boron	Reduced leaf area, curling of leaves, lamina deformation, appearance of white stripes perpendicular to the veins on the lamina of young leaves, thickening of secondary veins and inhibition of root and flower formation.	Foliar application of Borax @ 5 g/litre with good quality surfactant 2-3 times at monthly interval from 4 th month onwards is recommended.	
5.	Magnesium	Yellow discoloration in the mid of the leaf blade and midrib portion while the leaf margins remain green. Purple mottling of the petioles, marginal necrosis and separation of leaf sheaths from the pseudostem.	Application of Magnesium Sulphate (25g/plant) followed by irrigation	

6.	Sulphur	Yellow or white appearance of young leaves, necrotic patches on the leaf margins, thickening of veins, stunted growth and small or choked bunches.	Application of 30g of Bentonite Sulphur per plant at 3 rd or 5 th month after planting followed by irrigation is recommended.	
7.	Iron	The younger leaves turn yellow or white.	Foliar spraying Iron Sulphate (0.5%) along with 1% Urea is recommended.	
8.	Zinc	Young leaves become smaller in size and more lanceolate in shape. In the furling leaf high amount of anthocyanin pigmentation appear on its underside. The unfurled leaf has alternating chlorotic and green bands. Fruit is light green, twisted, short and thin.	Spraying Zinc Sulphate (0.5%) on the leaves is recommended	
9.	Manganese	Narrow green edge appears at the leaf margins of second or third youngest leaf which further spreads along the main veins towards the midrib. However, interveinal areas remains green giving comb tooth like appearance.	Foliar spraying of 0.5% MnSO_4 is recommended	

10.	Calcium	Deformation or absence of leaf lamina (spike leaf), marginal leaf necrosis and thickening of veins.	Application of lime or Dolomite @ 50g/plant followed by irrigation.	
11.	Copper	Both young and old leaves show symptoms of chlorosis and curve towards the base, which gives an umbrella like appearance to the plant.	Spraying Copper Sulphate (0.5%) on the leaves is recommended	

7. IRRIGATION MANAGEMENT

- The total water requirement of banana plants for its entire life cycle is about 900-1200 mm that can be met both through rainfall as well as supplementary irrigation in order to maintain optimum moisture at all stages of growth.
- Providing good drainage facilities to drain out excess water from the root zone equally important to promote better growth and enhance the productivity.
- In general irrigation of banana plantations every 3-4 days during hot period and 7-8 days' interval during cool weather is recommended.

Drip Irrigation

- Drip irrigation is superior to conventional basin irrigation in terms of water and nutrient use efficiency, more vigorous growth, higher yields and minimized weed growth.
- With about 30-40% saving of water and uniform application of inputs, drip system has the advantage of large area coverage with minimum requirement of water and labour.

Water requirement of banana at different growth stages under drip irrigation

S. No	Crop growth stage	Duration (weeks)	Quantity of water (lit./plant)
01	After planting / Ratoon	1-4	4-6
02	Juvenile phase	5-9	8-10
03	Critical growth stage	10-19	12
04	Flower bud differentiation stage	20-32	16-20
05	Shooting stage	33-37	20 and above*
06	Bunch development stage	38-50	20 and above*

* Depending on the season / Evapo-transpiration

8. INTERCULTURAL OPERATIONS:

8.1 Bunch covering:

This practice protects bunches against cold, sun scorching, against attack of thrips and scarring beetle. It also improves certain visual qualities of the fruits. Bunch covering with dry leaves is a common practice in India.



8.2 Deflowering:

It consists of removal of the withered style and perianth. They are to be removed just after flowering.



8.3 Denavelling:

The operation of removal of male buds after emergence of last hand of bunch is called 'Denavelling' which helps in fruit development and increases bunch weight.



8.4 De-suckering:

Removal of unwanted suckers is one of the most critical operations in banana cultivation and is known as desuckering. Such suckers are removed either by cutting them off or the heart may be destroyed without detaching the sucker from the parent plant. Removal of suckers with a portion of corm at an interval of 5-6 weeks hastened shooting and increased the yield.



8.5 Leaf Removing:

Pruning of surplus leaves helps to reduce the disease from spreading through old leaves. Leaf pruning can change light and temperature factors of microclimate. Pruning of leaves before bunch initiation delays flowering and harvesting cycle. For maximum yields a minimum of 10-12 leaves are to be retained.



8.6 Mattocking:

It is the process of cutting the pseudostem leaving a stump of about 0.6m height. The left over stump with its stored food material continues to nourish the daughter sucker (follower) till it withers and dries up. However, in areas of widespread occurrence of banana pseudostem borer and corm weevil, this practice is not encouraged. Instead, the harvested stem is chopped and mulched into the soil.



8.7 Propping:

Propping is an important operation to be carried out in areas with high wind speeds. Pseudostems are supported up with bamboo or casuarine or eucalyptus poles especially, at the time of bunch emergence.



8.8 Earthing Up:

In case of furrow planting earthing up should be done during rainy season to avoid water logging while during winter and summer the plant should be in the furrow.



9. Intercropping

Intercropping can easily be raised in banana plantation at the early stages of growth. Vegetable and flower crops like radishes, cauliflower, cabbage, spinach, chilli, brinjal, lady's finger, gourds, marigold, and tuberose can be successfully grown as intercrop. Mixed cropping with arecanut coconut and cassava is a common and widely adopted practice in South India. Avoid growing cucurbitaceous vegetables.

10. AGRO-ECOSYSTEM ANALYSIS BASED INTEGRATED PEST MANAGEMENT

The IPM has been evolving over the decades to address the deleterious impacts of synthetic chemical pesticides on environment ultimately affecting the interests of the farmers. The Economic Threshold Level (ETL) was the basis for several decades but in modern IPM (FAO 2002) emphasis is given to AESA where farmers take decisions based on larger range of field observations. The health of a plant is determined by its environment which includes physical factors (i.e. soil, rain, sunshine hours, wind etc.) and biological factors (i.e. pests, diseases and weeds). All these factors can play a role in the balance which exists between herbivore insects and

their natural enemies. Understanding the intricate interactions in an ecosystem can play a critical role in pest management.

Decision making in pest management requires a thorough analysis of the agro-ecosystem. Farmer has to learn how to observe the crop, how to analyze the field situation and how to make proper decisions for their crop management. This process is called the AESA. Participants of AESA will have to make a drawing on a large piece of paper (60 x 80 cm), to include all their observations. The advantage of using a drawing is that it forces the participants/farmers to observe closely and intensively. It is a focal point for the analysis and for the discussions that follow, and the drawing can be kept as a record.

AESA is an approach, which can be gainfully employed by extension functionaries and farmers to analyze the field situations with regards to pests, defenders, soil conditions, plant health and the influence of climatic factors and their relationship for growing a healthy crop.

The basic components of AESA are

- Plant health at different stages
- Built-in compensation abilities of plants
- Pest and defender population dynamics
- Soil conditions
- Climatic factors
- Farmers past experience

10.1 Principles of AESA based IPM:

10.1.1 Grow a healthy crop

- Select a variety resistant/tolerant to major pests.
- Select healthy seeds/seedlings/ planting material.
- Treat the seed/seedlings/planting material with recommended pesticides especially bio-pesticides.
- Follow proper spacing.
- Improve the soil health by mulching and green manuring, whenever applicable.
- Nutrient management especially with organic manures and bio-fertilizers based on the soil test results. If the dosage of nitrogenous fertilizers is too high

the crop becomes too succulent and therefore susceptible to insects and diseases. If the dosage is too low, the crop growth is retarded. Apply an adequate amount for nutrients for best results. The phosphatic fertilizers should not be applied each and every season as the residual phosphate of the previous season will be available for the current season also.

- Proper irrigation
- Crop rotation

10.1.2 Observe the field regularly (climatic factors, soil and biotic factors)

- Monitor the field situation at least once a week (soil, water, plants, pests, natural enemies, weather factors, etc.). Make decisions based on the field situation and Pest: Defender (P:D) Ratio. Take immediate action when needed (e.g. collection and destruction of egg masses, infested/infected parts/plants, etc).

10.1.3 Plant compensation ability

Compensation is defined as the replacement of plant biomass lost to herbivores and has been associated with increased photosynthetic rates and mobilization of stored resources from source organs to sinks (e.g., from roots and remaining leaves to new leaves) during active vegetative growth period. Plant tolerance to herbivore can arise from the interaction of a variety of plant traits and external environmental factors. Several studies have documented such compensation through increased growth and photosynthetic rate.

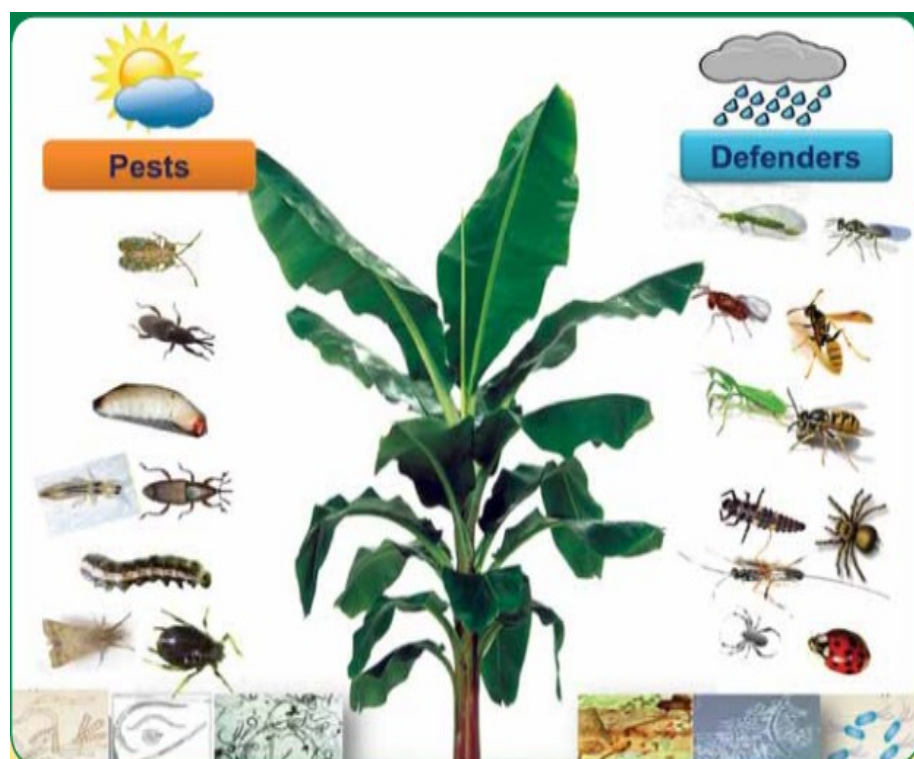
10.1.4 Understand and conserve defenders

- Know defenders/natural enemies to understand their role through regular observations of the agro-ecosystem.
- Avoid the use of chemical pesticides especially with broad-spectrum activity.

10.1.5 Pest: Defender (P: D) ratio:

Identifying the number of pests and beneficial insects helps the farmers to make appropriate pest management decisions. Sweep net, visual counts, etc. can be adopted to arrive at the number of pests and defenders. The P: D ratio can vary depending on the feeding potential of natural enemy as well as the type of pest. The natural enemies can be divided into three categories like 1. Parasitoids; 2. Predators;

and 3. Pathogens. The important natural enemies in mango are given in ecological engineering table.



10.1.6 Decision taken based on the analysis of field situations:

Soil conditions :
Weather conditions :
No. of insect pests
Diseases types and severity :
Weeds types and intensity :
Rodent damage (if any) :
No. of natural enemies :
P: D ratio :

The general rule to be adopted for management decisions relying on the P: D ratio is 2:1. However, some of the parasitoids and predators will be able to control more than one pest. Wherever specific P:D ratios are not found, it is safer to adopt the 2:1, as P: D ratio. Whenever the P: D ratio is found to be favourable, there is no need for adoption of other management strategies. In cases where the P: D ratio is found to be unfavourable, the farmers can be advised to resort to inundative release of

parasitoids/predators depending upon the type of pest. In addition to inundative release of parasitoids and predators, the usage of biopesticides and botanicals, etc. can be relied upon before resorting to synthetic chemical pesticides.

10.2. Field scouting:

Surveillance on pest occurrence at the main field should commence soon after crop establishment after transplanting and at weekly intervals thereafter. In each of the fields, select five spots randomly. Select five random plants at each spot for recording counts of insects as per procedure finalized for individual insects.

For sucking pests:

Aphids, whitefly: Count and record the number of both nymphs and adults on five randomly selected leaves per plant

Thrips: Count and record the number of nymphs and adults of thrips present on five terminal leaves per plant (tapping method also can be used to count thrips).

10.3 Root sampling:

Always check plants that appear unhealthy. If there are no obvious symptoms on plants, examine plants randomly and look for lesions or rots on roots and stems. Observe the signs of the causal organism (fungal growth). It is often necessary to wash the roots with water to examine them properly. If the roots are well developed, cut them to examine the roots for internal infections (discolouration & signs). Count the total number of pseudo stem infested/infected due to rot should be counted and incidence should be recorded.

10.4 Leaf Sampling:

Examine all leaves and or sheaths on each plant for lesions and determine the area of leaf infection. Count the number of leaves (leaf area diameter)/plant infected due to disease and record.

10.5 Stem, Flower And Fruit Sampling:

Carefully examine the stems, flower and fruits of plants for signs of fungal or bacterial diseases. The stems and fruits should be split or taken apart and examined for discoloration caused by fungi and bacteria. Count the number of plant, fruit, infected due to disease and record.

10.6 Monitoring Through Pheromone Trap Catches:

Pest specific pheromone traps should be used for lepidopteran pests for monitoring. Fix the traps to the supporting pole at a height of one foot above the plant canopy. Change the lures once in 2-3 weeks regularly. Number of moths/trap should be counted and recorded.

11. ECOLOGICAL ENGINEERING FOR PEST MANAGEMENT

Ecological engineering for pest management has recently emerged as a paradigm for considering pest management approaches that rely on the use of cultural techniques to effect habitat manipulation and to enhance biological control. The cultural practices are informed by ecological knowledge rather than on high technology approaches such as synthetic pesticides and genetically engineered crops (Gurr *et al.* 2004). Natural enemies may require:

1. Food in the form of pollen and nectar for adult natural enemies.
2. Shelter such as overwintering sites, moderate microclimate, etc.
3. Alternate host when primary host are not present

a. Ecological engineering for pest management – above ground:

- Raise the flowering plants / compatible cash crops along the field border by arranging shorter plants towards main crop and taller plants towards the border to attract natural enemies as well as to avoid immigrating pest population.
- Grow flowering plants on the internal bunds inside the field.
- Not to uproot weed plants those are growing naturally like *Tridax procumbens*, *Ageratum* sp, *Alternanthera* sp etc. which act as nectar source for natural enemies.
- Not to apply broad spectrum chemical pesticides, when the P: D ratio is favourable. The plant compensation ability should also be considered before applying chemical pesticides.

B. Ecological Engineering for Pest Management – Below Ground:

- Crop rotations with leguminous plants which enhance nitrogen content.
- Keep soils covered year-round with living vegetation and/or crop residue.
- Add organic matter in the form of farm yard manure (FYM), vermicompost, crop residue which enhance below ground biodiversity.
- Reduce tillage intensity so that hibernating natural enemies can be saved.

- Apply balanced dose of nutrients using biofertilizers.
- Apply mycorrhiza and plant growth promoting rhizobacteria (PGPR).
- Apply *Trichoderma viride/harzianum* and *Pseudomonas fluorescens* as suckers treatment and soil application (if commercial products are used, check for label claim).

However, for biopesticides produced by farmers for own consumption in their fields, registration is not required). More information on biodiversity of natural enemies and flowering plants that attract natural enemies / repel pests can be accessed by clicking the link <http://ppqs.gov.in/sites/default/files/banana.pdf>.

12. INTEGRATED PEST MANAGEMENT

12.1 Rhizome Weevil: *Cosmopolites sordidus* (Germar) (Coleoptera: Curculionidae)

Life cycle:

Egg: Elongate oval, about 2 to 3 mm long and white in colour. Eggs are laid singly in small cavities that are chewed out by the female at the base of the pseudostem just above ground level, in the upper part of the corm, in roots near the soil surface and at the end of cut stems (stumps). Egg period is 4-36 days depending on the temperature conditions.

Larva: Creamy white legless grubs, stout and distinctly curved and swollen in the middle of the body. The head is reddish brown with strong mouthparts. Fully-grown larvae are about 12 mm long. Larval period is 20 to 25 days.

Pupa: The pupa is creamy white and about 12 mm long. Pupation takes place in holes bored by the larvae. Pupal period is 5 to 7 days.

Adult: Adult weevil is 10-16 mm long, dark brown to dull black with a rather long curved snout. Newly emerged weevils are red brown, turning almost dark brown to black after a few days. They are free living, they are most commonly found in the soil at the base of the mat or associated with crop residues. The adult weevils may live for up to two years, and can live without food for six months, but are very sensitive to desiccation and will die within 48 hours if kept on a dry substrate. They are active at night.



Larva



Pupa



Adult

Nature of damage and symptoms

- Infestation at early stage reduces plant vigour and affected plants have a sickly appearance. Presence of yellow lines on the lamina is an early symptom.
- Extensive tunneling of the larvae in the corm weakens the plants, making the affected plants prone to wind damage
- In advanced stages of infestation, plants show tapering of stem at crown region, reduction in leaf size and number, poor bunch formation and choked throat appearance of bunch due to grub damage on corm
- The adult weevil infests at the crown region where the corm and stem join. Yield losses exceed 50% in ratoon crops.



Management

- Use healthy and pest free suckers for planting.
- Clean and weed free cultivation is essential to avoid the spread of weevils.
- Removal of roots and outer skin of corms and dipping them in solution of *Beauveria bassiana* (1×10^9 CFU/ml) for about 20 minutes before planting helps to kill the eggs and grubs of the corm weevil and plant parasitic nematode.
- Weevils are attracted to their hosts by volatiles, especially following damage to the rhizome and their activity can be monitored by keeping longitudinal split banana pseudostem traps at 10-15/ha. On noticing weevil catches on these traps, 100 traps / ha may be kept in the field with their cut ends smeared with biocontrol agents like liquid formulation of *Beauveria bassiana* (1×10^9 per ml.) or 20g rice chaffy grains formulation of *Beauveria bassiana* or 20 ml of entomopathogenic nematode, *Heterorhabditis indica*. The stem traps need to be placed with the cut surfaces facing the ground. Males produce an aggregation pheromone that is attractive to both sexes.
- Uproot and destroy infested rhizomes and pseudostems after the harvest and treat with *B. bassiana* liquid formulation as above to kill the weevils.

12.2 Pseudostem Borer: *Odoiporus longicollis* (Oliver) (Coleoptera: Curculionidae)

Life cycle:

Egg: Eggs are cream in colour and cylindrical in shape with rounded ends. The incubation period ranges from 3 to 8 days.

Larva: The emerging larvae are fleshy, yellowish white and apodous. The larvae feed on tissues of the succulent sheath by tunneling extensively and may reach as far as the true stem.

Pupa: Pupation takes place in cocoons made by winding short pieces of fibrous materials of the sheath around its body. The pupa is exarate and present inside the cocoon.

Adult: The adult weevils are black and measure 23-39 mm. They often confine themselves within the pseudostem and in the decomposing tissues of harvested pseudostems. Adults live up to a year.



Larva



Adult

Nature of damage and symptoms:

Infestation of the weevil normally starts in 5-month-old plants. Early symptoms of the infestation are (a) presence of small pinhead-sized holes on the stem; (b) fibrous extrusions from bases of leaf petioles; and (c) presence of adult weevils and exudation of a gummy substance from the holes on the pseudostem. Larvae bore in the pseudostems and bunch stalks. The long larval tunnels rapidly become discoloured. If the larval tunnel passes through the central growing point, the plant is killed. During the advanced stages of infestation, extensive tunneling is observed both in the leaf sheath and in the pseudostem. Rotting occurs due to secondary infection of pathogens and a foul odour is emitted.



Holes on banana stem covered with slimy, jelly like exudate



Discoloured tunnels on the affected stem



Banana split stem trap

Management:

- Stem weevil activity in a garden can be monitored by keeping the banana longitudinal split (upside down) traps and if the weevils are attracted to the traps, then control measures must be taken up immediately.
- Follow clean cultivation practices and in severe infestations, uproot and burn the plants. Maintain healthy plantation by periodical removal of dry leaves and suckers. Prune the side suckers every month

- Apply mud slurry mixed with neem oil 5% on the pseudostem five months after planting in hotspots to prevent oviposition
- Do not dump infested materials into manure pits after the harvest of banana bunch - cut the tree at the base and treat with 100 ml of *Beauveria bassiana* (3 ml/litre).
- Stem trapping - Cut old pseudostems into lengths of 20-60 cm and split them open longitudinally, place on the ground near the crown bases with the cut surface downwards. Swab the cut surface of these traps with biocontrol agents like *B. bassiana* (20 g rice chaffy grains formulation/ 3 ml in 100 ml of liquid formulation) or 15 ml of entomopathogenic nematode, *Heterorhabditis indica* at 1 billion infective juveniles (IJs) per ml. The volatiles from these stem traps attract weevils that will be killed while walking on the trap surface within or after 24 hrs. *Metarhizium anisopliae* can also be used in place of *B. bassiana*.

12.3 Banana Aphid: *Pentalonia nigronervosa* (Coquerel) (Hemiptera: Aphididae)

It has been implicated in the transmission of banana bunchy top virus, bract mosaic virus and 3 plant viruses.

Life Cycle:

- Adult aphids have characteristic wings with dark brown to black venation. Aphids live in colonies within leaf axils or tender leaves and beneath the leaf sheath at base of the stem at the ground level.
- Nymphs undergo four instars lasting 8-12 days.
- Adults reproduce parthenogenetically 35-50 nymphs during a life span of 27-37 days.
- Life cycle completed in 12-15 days, about 30-40 overlapping generations are completed in a year.

Nature of damage and damaging symptoms

- Aphids are seen in colonies on leaf axils and pseudostems. Dark reddish brown to black, shiny wingless aphids can be seen on the young, emerging suckers. The aphid causes indirect damage by transmitting the virus disease

bunchy top. Green streaks initially appear on the secondary veins on the ventral side of the lamina. The affected leaves become brittle, small and petioles get elongated. Crown composed of narrow stunted leaves gives a bunchy top appearance.



Aphid colonies on leaf axils



Nymph



Adult

Management

- Spray *Verticillium lecanii* @ 20 g or 5 ml /l at 25, 65 and 165 days after planting.
- Apply neem cake @ 1 kg in two equal splits at one and two months after planting.
- Spray 2 % neem oil garlic emulsion directing the spray on leaf axil and base of the plant.
- Predators of banana aphids include ladybird beetles (*Scymnus nubilus*, *Pseudaspidimerus trinotatus*, syrphids (*Paragus serratus*) and a hemerobiid (*Micromus timidus*). Protect these natural enemies.
- Spray Dimethoate 30 EC @ 1.485-1.980 lit/ha or Oxydemeton methyl 25%EC @ 3-4 lit/ha.
- Spray soapy water or insecticidal soap on plants thoroughly on petioles, furred leaves, whorls or on young suckers

12.4 Lacewing Bugs: *Stephanitis typicus* (Distant) (Hemiptera: Tingidae)

Life Cycle:

Adults are pale with glassy, transparent wings. Colonies of nymphs and adults are found on the lower surface and suck the sap.

Nature of damage and symptoms:

The feeding causes yellowing of leaf on the upper side. Feeding causes small white spots to appear on the upper leaf surface opposite the feeding site. Chlorotic spots and dark excreta marks are left on the lower leaf surface.



Management:

- Use yellow sticky trap at 15/ha
- *Stethoconus praefectus* (Distant), an obligate predator of *S. typica*, is commonly found on infested leaves. Two egg parasitoids, *Erythmelus panis* Enock and *Anagrus* sp. (Hymenoptera: Mymaridae) are reported as egg parasitoids of *S. typica* from Tamil Nadu and Manipur, respectively.
- Spray dimethoate 30 EC @ 1.485-1.980 lit/ha or oxydemeton methyl 25 EC @ 1.5-2.0 lit/ha or quinalphos 25 EC @ 3-4 lit/ha.

12.5 Thrips: Leaf thrips (*Helionothrips kadaliphilus*), Flower thrips (*Thrips hawaiiensis*) (Thysanoptera: Thripidae)

Life cycle:

Leaf thrips: It has reddish brown body with yellowish grey wings. It is found congregating on the ventral surface of the leaves and sucks the sap. The feeding spots turn yellowish brown and in severe cases the affected leaves dry up.

Flower thrips: Adults and nymphs feed on flower tips and fruits before the bud emerges from the plant and it also persists up to two weeks of emergence. Due to

sucking by the adult and nymphal stage, the fruits develop black spots (pics). Poovan, Nendran and Karpooravalli are susceptible.



Egg

Larva

Pupa

Thrips

Nature of damage and symptoms:

Both leaf thrips and flower thrips cause fruit damage. The early symptoms appear as water-soaked smoky areas where the colonies congregate to feed and oviposit between touching or adjacent fruit. These areas then develop the typical rusty-red to dark brown-black discolouration. Further rusty growth of the fruit and yellowing of leaves. While the taste and texture of the fruit within these peels remains unaffected, the exterior discoloration reduces the marketability of affected fruit. In severe cases the skin develops longitudinal cracks and sometimes fruit may split.



Red rusty spots on fruits



Red rusty streaks on fruit skin



Fruit cracking



Yellowing of leaves

Management:

- Regular checking of fruit under the bunch covers is essential to ensure that damage is not occurring.
- Use blue pan water sticky trap @ 4-5/acre.
- Use thrips-free planting material or tissue-cultured bananas and, if possible, hot water treatment prior to planting.

- Destroy all the volunteer plants and that could act as a source of thrips to spread to other plantings.
- Bunch sleeving at the time of shooting with polypropylene covers (which cover the full length of the bunch) do provide some protection on the developing fingers.
- To kill the thrips pupa in the soil, apply liquid formulation of *Beauveria bassiana* (1×10^9 Spores/ml).

12.6 Rugose Spiraling Whitefly: *Aleurodicus rugioperculatus* Martin (Hemiptera: Sternorrhyncha: Aleyrodidae)

The rugose spiraling whitefly is an invasive pest that attacks a wide range of host plants including palms, coconut and banana are among the most preferred host plants. Severe infestation of this pest was observed on banana besides coconut.

Life Cycle:

The eggs are laid in a spiralling pattern and have a very short stalk. The emerging nymph is yellowish and develops white waxy filaments later. The adult whitefly, though, is quite different from the spiralling whitefly by its much larger size and in having two pale brown wavy markings on the forewings, one medial and one apical.



Eggs on Banana leaf



Rugose Spiralling Whiteflies on Banana



Nature of damage and symptoms:

Nymphs and adults of the whitefly suck the sap on the under surfaces of the leaf. Extensive feeding of the insect leads to the excretion of honeydew which subsequently gets deposited on the upper surface of the leave. Honey dew excrement, being sweet and watery, attracts ants and encourages growth of the fungus *Capnodium* sp. which causes disfigurement of hosts affecting the photosynthetic

efficiency of the plant. In some cases, complete drying of leaves was observed on banana in some places.



Sooty Mould Deposition on the Rugose Whitefly infested Banana

Management:

- *Encarsia guadeloupae*, an exotic parasitoid of the spiralling whitefly introduced in India in 1999, causes 60–70% parasitism of this whitefly.
- Indiscriminate use of insecticides may be avoided so that these parasitoids and other indigenous natural enemies are not adversely affected.
- Spraying starch solution (1%) to dislodge the heavy sooty mould deposition on the leaves of infested plants.
- Use of yellow sticky traps to trap the adult whiteflies
- In case of severe infestation, spray neem oil 0.5%.

12.7 Nematodes

Lesion forming species

- Burrowing nematode: *Radopholus similis*
- Root-lesion nematode: *Pratylenchus coffeae*
- Spiral nematode: *Helicotylenchus multicinctus*

Gall forming species

- Root knot nematode: *Meloidogyne incognita* & *M. javanica*

Nature of damage and symptoms

Lesion forming nematodes causes reddish-brown to black, elongated lesions which are readily seen when the roots are split open. Roots eventually blacken and die. Attacks by nematodes, combined with the effects of secondary micro-organisms,

destroy or weaken much of the root system. Infested plants lack vigour and fruiting is poor. Such plants toppled during heavy wind.



Affected plant



Toppling of affected plant



Affected root system



Lesions on corm



**Lesions on cord roots
(Longitudinally split)**



Roots damaged by root-knot nematode

Identification of nematodes

a. Spiral nematode (*Helicotylenchus multicinctus*)

Spiral Nematodes are robust with strong stylets. This is an ecto as well as endo – parasite. They live partially or completely within roots, feeding on the outer cortical cells. The nematode causes superficial lesions on the roots. The development of feeder roots is affected in case of high population of nematodes in the soil.

b. Root lesion nematode (*Pratylenchus coffeae*)

These are migratory endoparasites. In case of severe infestation, the cortical region of the root is completely damaged and turns black in colour. Life cycle takes less than 20 days when temperatures range from 25°C – 30°C

c. Root knot nematodes (*Meloidogyne incognita* & *M. javanica*)

They are sedentary endoparasites. They enter the roots remain inside and complete their life cycles. The infected roots exhibit characteristic swellings and galls. In general the nematode affected plants exhibit pale yellow leaves with dried margins.

In case of severe infestation, the plants may fall off on slight push by hand or wind. Planting of infested suckers do not establish and delay growth. Chocking of bunches, reduction in total weight and size of fruits are also encountered as a result of heavy infestation.

Management:

- Plant suckers obtained from nematode free / uninfected land
- Pare / trim the suckers until all black or discolored spots have been removed, leaving only clean white tissues.
- Submerge trimmed suckers for 20 minutes in hot water at 50-55°C.
- Grow nematode resistant cultivars.
- Well decomposed manure should be used.
- Crop rotation with non-host crops.
- Grow nematode antagonistic crop such as marigold as an intercrop.
- Covering the field for 6 to 8 weeks with plastic mulch after tilling and watering for soil solarization.
- At the time of planting the apply neem cake @ 500g per pit.
- For control of *Meloidogyne incognita*, apply 2 kg of *Trichoderma harzianum* 1.5% WP (Strain No. IIHR-TV-5 Accessions No. ITCC 6889) enriched FYM @ 2 kg/plant at the time of planting and at an interval of 3 months after planting for a period of one year.

12.8 Burrowing nematode/Toppling over/Black head disease: *Radopholus similis*

It is a migratory endoparasite. The nematode enters the roots as well as corms and causes usually induce reddish brown cortical lesions which are characteristic feature of the disease. These lesions are clearly seen when an infected root is split longitudinally. Root and rhizome necrosis is manifested by the retarded growth, leaf yellowing and falling of mature plants.

The lessening of primary roots together with girdling and death of anchoring roots leads the plant prone to 'tip over' under wind pressure. Burrowing nematode laid the eggs in the corms and roots. 2 weeks are required for the life cycle to be

completed. Female nematodes live 2 to 3 months and lay more than 100 eggs each. Nematode survive in the soil in the absence of suitable hosts.



Fallen trees due to severe infestation

Reddish brown discoloration on infected rhizomes

Mode of Spread and Survival:

- Migratory endoparasite and obligate in nature.
- Primary source of inoculum: Infected soil and suckers.
- Secondary source of inoculum: Soil - nematodes (J-2 stages).
- Use disease free suckers.

Management:

- Avoid excess nitrogen application in the soil.
- Apply 5-10 kg of FYM at the root zone per plant.
- Application of Neem cake with FYM @ 5kg/plant has also been found effective.
- If nematode population is very high in the field, crop rotation with paddy, sugarcane, green gram, sun hemp or cotton is recommended to reduce the nematode population.
- Fallowing the land for three months after banana harvest and summer ploughing effectively suppress the nematode population.
- Hot water treatment of suckers at 55° C for 15-25min.
- Apply neem cake@500gm/plant at the time of planting and at four months after planting with any one of the biocontrol agents *Trichoderma harzianum* 1.5% WP @ 20 gm/plant at planting and 3rd and 6th month after planting.
- Grow marigold as intercrop in banana.
- Grow tolerant banana varieties like Karpuravalli, Monthan, Nattupooovan, Kunnan etc.

13. INTEGRATED DISEASE MANAGEMENT

13.1 Panama wilt/Fusarium wilt: *Fusarium oxysporum* f. sp. *cubense*

Symptoms

- Symptoms are most clear on the plants that are at least five-month-old.
- Yellowing of the lower most leaves starting from margin to midrib of the leaves and spreads to upper leaves.
- Drooping- leaves break near the base and hang around stem gives skirt like appearance.
- Stem splitting- Longitudinal splitting of pseudo stem.
- Discolouration of vascular vessels as red or brown streaks on periphery of pseudostem.
- Wilting of the plant due to blocking of xylem and phloem vessels.
- Race 1, Race 2 and Race 4 are reported to be present in India attacking most of the banana varieties including Cavendish bananas



Splitting of pseudo stem



Drooping & yellowing of leaves



Discolouration of vascular vessels

Disease Development:

- The pathogen is soil borne.
- Primary source of Inoculum: Chlamydospores in the soil and Infected rhizomes
- Secondary spread: Irrigation water and soil adhering to agriculture implements.

Continuous cultivation results in build-up of inoculum in soil

Survival: As chlamydospores in soil and in infected suckers used for planting.

Dispersal and Aggravation:

- Micro and macro conidia through irrigation water.
- Severe in acidic and light textured soils
- Low soil moisture, Flooding irrigation , Mono-cropping
- Cultivation of banana in poorly drained heavy soils.
- Cultivation of susceptible cultivars
- Infection by nematodes predisposes the plants to disease.

Management:

- Use of disease-free suckers, tissue culture plants for planting instead of suckers when the soil is not known to have the history of Fusarium wilt.
- Removal of infected plants and application of lime @1-2 Kg/pit.
- Crop rotation with paddy, Sugarcane, Tapioca, Onion, Pineapple.
- Before planting banana, the field must be inundated/flooded with water for a period of 1 to 3 months, biological disinfection method may be followed i.e. spreading 500 to 1000kg/acre of paddy or maize straw and flooding the field for 20 to 30 days, Sun hemp seeds may be broadcasted, grown for 45 days and *in situ* ploughing may be carried out.

13.2 Eumusae Leaf Spot Disease *Mycosphaerella eumusae*

Symptoms:

- On leaves small light yellow or brownish green narrow specks appear.
- They enlarge in size becomes linear, oblong, brown to black spots with dark brown border and yellow halo.
- Black specks of fungal fruitification appear in the affected leaves.
- Rapid drying and defoliation of the leaves.
- Reduces photosynthesis, size of bunches and fruit size.



Brown to black spots on



Black specks of fungal



Rapid drying of leaves

the leaves

fructification

Disease Development:

- The fungus survives in the infected banana leaves
- Secondary infection is by the windborne conidia of the fungus.
- Surface moisture is necessary for release of both conidia and, hence the disease is severe in moist weather.
- High humidity, heavy dew and rainy weather.
- Soils with poor drainage and low fertility favour the disease.
- Thick planting and presence of weeds promote disease development.

Management:

- Planting banana in well drained soils.
- Remove affected leaves and burn.
- Planting at recommended density.
- Pruning suckers periodically to avoid overcrowding in the field.
- Applying recommended dose of fertilizers.
- Foliar spray of *Bacillus subtilis* @ 5 lit/ha.
- Spray Propiconazole @ 1 ml / litre of water for 3 to 5 times at 25 to 30 days interval.

13.3 Anthracnose: *Colletotrichum musae*

Symptoms:

- Blister like spots appear on green fruit and ripened fruits and distal end of the hands.
- These blister spots turn to dark brown or black colour.
- The fungus produces masses of conidia which form a pinkish coat.
- The entire fruit and bunch are affected in severe case.
- Premature dropping of infected fruits.
- Presence of Latent infection- “Infection occurs in the green stage and expression of symptoms on ripened stage”



Shriveling of affected fruits



Black spots on fruits

- Optimum temperature - 30-35°C
- Relative humidity of above - 85%
- The incidence is severe during June-September when the temperature is high.
- The spread of the disease is by air borne conidia and insects which frequently visit banana flowers.

Management:

- Avoid damage to fruits at harvest and transit.
- Removal of all the dead hanging leaves at regular intervals. Sleeve the bunches as soon as possible after the removal of male bud. Burn the infected materials.
- Proper field sanitation. Keep the field free of weeds and provide good drainage.
- Fruit should be free from infection before it is transported, stored and ripened.
- Banana bunches should be harvested at correct stage of maturity.
- Proper fertilization prevents the infection.
- Spray Copper Oxychloride 50 % WP @ 2.5 gram/ml.

13.4 Cigar end rot: *Verticillium theobromae*

Symptoms:

- A black necrosis spread from the perianth into the tip of immature fingers.
- The pulp of the affected fruit undergoes a dry rot.
- The dry rot can spread up to 2 cm from the tip.
- The necrotic tissue become covered with the fungal mycelia and resembles the “greyish ash of the cigar-end”.
- If emerging fingers are affected, the entire fingers sometimes get rotten.



Black necrosis starts from tip



Necrotic tissue covered with mycelia

Disease Development:

- The disease incidence is high during summer season.
- The fungus is commonly seen in leaf trash and flowers.
- Conidia are disseminated in air currents and infect flower parts.

Management:

- Field sanitation - removal of infected plants.
- The plantations should have enough aeration
- After emergence, the bunches are covered with polyethylene bag.
- Packing and ripening room need to be kept clean.
- Spray of Mancozeb 75% WP @ 1.5-2 kg/ha.

13.5 Tip over/Rizome rot : *Ervinia caratovora* sub sp. *caratovora*

Symptoms:

- This disease is more pronounced on young suckers leading to rotting and emitting of foul odour.
- Rotting of collar region followed by epinasty of leaves, which dry out suddenly.
- If affected plants are pulled out, it comes out from the collar region leaving the corm with their roots in the soil.
- Pseudostem easily come out from corm portion on exerts slight pressure.
- Oozing visible on the edge of corm and pseudostem confirm bacterial infection.



Crown rotting



Soft rotting of suckers



Drying leaves

Disease Development:

- The pathogen is a Gram-negative bacterium with peritrichous flagella and it is a rod-shaped bacterium that lives alone or aggregates into pairs and chains
- The pathogen is soil borne and enters through wounds and also through leaf sheath of suckers.
- The disease can be spread by infected plant debris, plant wounds and injuries. Hot and damp weather with plenty of rainfall trigger the disease to occur. Water is required for the bacteria to invade into the plant bacteria spread in irrigation water.

Management:

- Good drainage and soil conditioning can control the disease to some extent.
- Plant disease free suckers.
- Remove infected plants immediately.
- Remove plant residues after harvest.
- Practice crop rotation by using crops that are not susceptible to the bacterial rot disease like soybean, forage legumes and small grains. Banana should not be grown along with onion and other vegetables.
- Control nematodes and other insect pests that serve as vectors of the bacteria to invade the plant tissues.
- Use of rhizomes with dead central buds and active lateral buds prevents the appearance of the disease.
- Avoid planting during rainy season and use of bigger suckers (more than 500g) for planting.
- Drench with Sodium hypochlorite 10% or Bleaching powder 10g /litre/tree.

- Drench suckers 2% bleaching powder to control the disease in field at planting followed by another drenching the soil after 3rd month of planting to control the pathogen.

13.6 Bunchy top: Banana bunchy top virus

Symptoms:

- Dark broken bands of green tissues on the veins, leaves and petioles.
- Plants are extremely stunted.
- Leaves are reduced in size, marginal chlorosis and curling.
- Leaves upright and become brittle.
- Many leaves are crowded at the top, resembles bunchy top appearance.
- Branches size will very small, if infected earlier no bunch will be produced.



Reduced leaf size



Bunchy appearance of leaves



Crowded leaves at the top

Disease Development:

- The disease is transmitted primarily by infected suckers.
- Secondary spread is through the Aphid vector *Pentalonia nigronervosa* in persistent manner.

Management:

- Adaptation of strict quarantine measures.
- Eradication of all infected suckers.
- Certified virus free suckers (Tissue culture) should be used for planting.
- Remove and rouging of infected banana plants
- Maintain clean, weed free field for early detection of infested suckers
- Apply Carbofuran 3C G @ 166 g /plant or spray Dimethoate 30EC @ 1 ml/l to control the vector.

- The spray may be directed towards crown and pseudostem base up to ground level at 21 days interval at least thrice.

13.7 Banana bract mosaic disease: *Banana bract mosaic virus* (BBMV; potyvirus)

Symptoms

- The infected plants flower but produce very small bunches with curved brittle fruits.
- Very severely affected plants may fail to flower and may die by stunted growth and necrosis of pseudostem.
- The male buds are dark purple in colour with mosaic patches.



Reddish streaks on pseudostem



Mosaic streak on bracts



Disease Development

- The disease is caused by a virus belonging to potyvirus group. The virions are flexuous filamentous
- The virus is transmitted through aphid vectors such as *Aphis gossypii*, *Pentolonia nigronervosa* and *Rhopalosiphum maidis*. In field the disease spread mainly through suckers

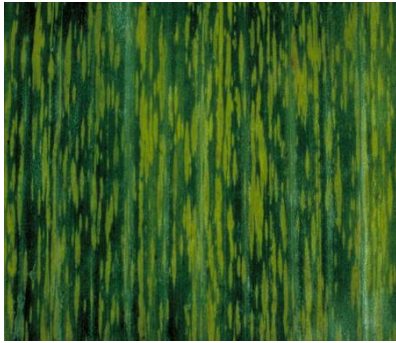
Management

Refer management given for CMV

13.8 Banana Streak Disease - *Banana streak virus* (BSV)

Symptoms

- A prominent symptom exhibited by BSV is yellow streaking of the leaves, which becomes progressively necrotic producing a black streaked appearance in older leaves.



Yellow streaks on leaves



Black streaks on older leaves

- The virus is transmitted mostly through infected planting materials, though mealy bugs (*Planococcus citri*) and more probably *Saccharicoccus sacchari* are also believed to transmit it. Shoot tip culture does not eliminate it from vegetatively propagated materials.

Management

- Use only virus-free planting material, taking it from plantations that are free from banana streak disease.
- Regularly monitor plantations and remove diseased plants as soon as they are seen. Take them out of the plantation and burn them. Replace with disease-free plants.

13.9 Infectious Chlorosis: *Cucumber mosaic virus* (Cucumovirus)

Symptoms

- The disease is characterized by the presence of typical mosaic-like or discontinuous linear streaking in bands extending from margin to midrib.
- Rolling of leaf margins, twisting and bunching of leaves at the crown and a rigid erectness in newly emerged leaves

- The presence of dead or drying suckers is noticed in advanced cases referred as heart rot resulting from rotting of heart leaf and central portion of pseudostem
- Primarily infected banana plants develop severe mosaic symptoms in young growth showing broadly streaked chlorotic or yellowish green bands and patches or chlorotic mottling distributed in patches over the leaf lamina
- The leaves are narrower and smaller than normal and the infected plants are dwarf and lag behind in growth. Such plants do not produce bunches but as a virus reservoir.



Disease development

- The causal agent of this disease is Cucumber mosaic virus (CMV). The virus is isometric, linear positive sense and single stranded RNA.
- The primary transmission is through use of infected daughter suckers from diseased plants and the secondary spread of the disease is through melon aphid, *Aphis gossypii* and *Aphids maidis*.

Management

- The banana gardens should be kept free from weeds
- Use virus indexed certified tissue culture plant.
- Infected suckers should not be used for planting
- Growing pumpkin, cucumber and other cucurbits between the rows of banana crop should be avoided
- Early detection by regular inspection of planting and eradication of diseased plants from the field as soon as they are noticed
- Use of high input crop management of system comprising of 10 kg farm yard manure at the time of planting and subsequently at an interval of 3 months. 1

kg neem cake, 200 gm nitrogen, 40 gm phosphorus, 200 gm potassium per plant, 4 weeding at 2months interval up to 8months stage.

- Spraying Dimethoate 30EC @ 1 ml/l or Methyl demeton 25 EC @ 2 ml/l at 3 to 4 weeks' interval controls the vector and reduces the spread of the disease.

14. Weed management

- Regular weeding is important during the first four months. Spading is commonly used and normally four spadings a year are effective in controlling weeds. Integrated weed management by including cover crops, judicious use of herbicides, intercropping and hand weeding wherever necessary will contribute in increased production. Pre-emergence application of Diuron @ 2kg /ha effective in controlling grasses and broad-leaved weeds without affecting the yield and quality of banana.
- Double cropping of cowpea is equally effective in suppressing the weed growth. Intercropping can easily be raised in banana plantation at the early stages of growth. Vegetable and flower crops like radishes, cauliflower, cabbage, spinach, chilli, brinjal, lady's finger, gourds, marigold, and tuberose can be successfully grown as intercrop. Mixed cropping with arecanut coconut and cassava is a common and widely adopted practice in South India.

15. DO'S AND DON'TS IN IPM:

S. No.	Do's	Don'ts
1.	Grow only recommended varieties.	Do not grow susceptible varieties.
2.	Transplant early in the season.	Avoid late sowing as this may lead to reduced yields and incidence of white grubs and diseases.
3.	Use disease free certified rhizomes. Always treat the rhizomes with approved chemicals/ bio products for the control of seed borne diseases/pests.	Do not use rhizomes without treatment with biocides/chemicals.
4.	Transplant in rows at optimum depths under proper moisture conditions for better establishment.	Do not sow rhizome beyond 15-20 cm depth depending on the root length.
5.	Apply only recommended herbicides at recommended dose, proper time, as appropriate spray	Pre-emergent as well as soil incorporated herbicides should not be applied in dry soils. Do not apply

	solution with standard equipment along with fl at fan or fl at jet nozzles.	herbicides along with irrigation water or by mixing with soil, sand or urea.
6.	Maintain optimum and healthy crop stand which would be capable of competing with weeds at a critical stage of crop weed competition.	Crops should not be exposed to moisture deficit stress at their critical growth stages.
7.	Judicious use of NPK fertilizers as per the soil test recommendation.	Avoid imbalanced use of fertilizers.
8.	Use micronutrient mixture after planting based on recommendations.	Do not apply any micronutrient mixture after planting without recommendations.
9.	Conduct AESA weekly in the morning preferably before 9 a.m. Take decision on management practice based on AESA and P: D ratio only.	Do not take any management decision without considering AESA and P: D ratio.
10.	Install pheromone traps at appropriate period.	Do not store the pheromone lures at normal room temperature (keep them in refrigerator).
11.	Spray pesticides thoroughly to treat the under surface of the leaves, particularly for scales, aphids, etc.	Do not spray pesticides only on the upper surface of leaves.
12.	Apply short persistent pesticides to avoid pesticide residue in the soil and produce.	Do not apply pesticides during preceding 7 days before harvest.
13.	Follow the recommended procedure of trap or border crops technology.	Do not apply long persistent pesticides on trap crop, otherwise it may not attract the pests and natural enemies.

16. POST HARVEST HANDLING

a. Curing and washing: During the curing, field heat of the fruit is brought down; this helps in stabilizing the metabolic process. The fruits are spread on the floor in orchard's yard, having the cushion of paddy straw and then washed to remove the dirt. By washing the original colour and lustre of the fruits is also recovered.

b. Ripening: Mature fruits are ripened with lower doses of Ethel for uniform colour development (slow ripening under controlled condition at 15⁰ -18⁰ C).

c. Grading: During grading, the produce is sorted according to the fixed grade standard, taking into consideration various quality factors to make a homogenous lot. Immature, overripe, damaged and diseased fruits are discarded during the process of grading.

d. Packaging: Wooden or cardboard boxes, rectangular in shape and bamboo baskets are used for packaging and transportation of fruits. In some cases, banana bunches are packed in old gunny bags wrapped with banana leaves. Firstly, fingers are removed from the bunch and washed in water. Then they are washed in dilute sodium hydrochloride solution to remove the latex, and finally air dried. These fingers are graded on the basis of their length & girth and packed in plastic corrugated fiber board (CFB) cartons having capacity of about 13 to 14.5 Kg. A suitable packing material like foam etc. may be used. These boxes are kept at 13-15⁰ C temperature and at 80-90% humidity in the cold storage. Bananas can be stored in such controlled atmosphere in a cooling chamber for a period of 20-25 days.

e. Pre-cooling: Precooling of the produce should be done within 10-12hrs of bunch harvesting. The fruit packed in boxes should be pre-cooled by forced air cooling at 13°C and 85-90% RH. It may take 6 to 8hrs to bring the fruit pulp temperature to 13°C from field temperature of 30° C to 35°C. The boxes should be immediately moved to cold rooms for storage purpose.

f. Storage: Bananas can be exported successfully by sea-shipment if the guidelines related to harvest maturity are strictly followed. Storage conditions of 13°C and 85% to 95% relative humidity are required. Storage temperature below 13°C would cause chilling injury to fruits resulting in surface dis-colouration, dull colour, failure to ripen and browning of flesh. The storage life at 13°C depends on the cultivar and varies from 3 to 4 weeks. A combination of low temperature with controlled atmosphere storage can further extend the storage life. Banana (Robusta cvr.) fruits could be stored in green unripe condition for 8 weeks under controlled atmosphere storage condition of 5% O₂ + 5% CO₂ at 12°C to 13°C with post storage ripening period of 4 to 5 days under ambient conditions.

g. Transportation: It is one of the critical stages in the banana supply chain. Poor transport conditions, rough handling, and delays in transportation contribute to losses

in banana supply chains. Bananas are transported from the field to the collection site and from the collection site to the retail market.

17. EXPORT PROCEDURES

Banana is an important fruit crop of many tropical and subtropical regions in India. It is the fourth largest contributor in total export of fruits from India. Amongst fruits, India ranks first in production of bananas 25.7 per cent (APEDA). Major banana producing states are Andhra Pradesh, Tamil Nadu, Maharashtra, Gujarat, Kerala and Karnataka. Among which Maharashtra stood as the largest banana exporting state with 96.91 per cent share in value. The major destinations for Indian bananas are UAE, Saudi Arabia, Oman, Iran and Bahrain respectively.

The pack house is to encourage exporters of horticultural produce to meet the international standards in terms of quality of produce with quarantine safety. The pack house development as per the requirement for material handling, holding, pre-inspection, shorting, grading, washing, standard treatment, packaging, stacking, pre-cooling, cold storage, quarantine check etc., encouragement to backward linkages, maintain the traceability, export of good quality produce, promotion and publicity at the international level, encouraging adoption of internationally acceptable standards and practices, up-gradation of standards for product quality, hygiene, food safety, and creation of a healthy, competitive environment among exporters. The various stakeholders are involved and getting the responsibilities for compliance of procedural formalities for quality export as per the importing countries. For more details visit the websites: Directorate of Plant Protection, Quarantine & Storage website: <http://plantquarantineindia.nic.in>, The Agricultural and Processed Food Products Export Development Authority (APEDA) <https://apeda.gov.in/apedawebsite/>, Banana Export system TNAU.

18. CIB&RC approved/ recommended pesticides for banana

Major uses of bio-insecticides					
Trichoderma harzianum 1.0% WP (Strain No. IIHR-TH-2 Accessions No. ITCC 6888)					
Name of crop	Name of insect				
Banana	Meloidogyne incognita	Apply 2 Kg Nemastin enriched FYM @ 2 Kg/plant at the time of planting and at 3 months interval for a period of one year			
Plant growth regulators (PGR)					
Gibberellic acid 0.001%L					
Name of PGR & approved crops	Time of application / purpose	Dosage /ha		Dilution in water (litres) / preparation of solution	Waiting period / PHI between last application & harvest (days)
		a.i. (ppm/gm/%)	Formulation (ml/gm/L tr/kg/%)		
Banana	a) First spray 3 rd month b) Second spray 5 th month c) Third spray at the time of fruit formation	0.027gm	270 ml.	450-500	-
Herbicides					
Diuron 80% WP					
Herbicide name & approved crops	Weed species	Dosage /ha		Dilution in water (litres)	Waiting period / PHI between last application & harvest (days)
		a.i. (gm/ Kg)	Formulation in (gm/ml /Kg/ ltr)		
Banana	Cyperus iria, Commelina benghalensis, Digitaria sp, Amaranthus spp, Dactyloctenium aegyptium, Chloris barbata, Eragrostis zeylenica,	1.60 Kg	2 Kg	625	-
Major uses of bio-fungicides					
Name of crop	Common name of the disease	Dose/ha		Dilution in water (litre/ha)	Waiting period a.i. (Days)
		a.i. (Days)	Formulation (g/ml)/%		
Bacillus subtilis 1.50% LF (T stanes BS-1 strain MTCC 25072)					
Banana	Eumusae leaf spot (Mycosphaerella eumusae)	-	5 litre/ha (Foliar spray)	750 Litre/ha	-

Bacillus subtilis 1.50% LF (T stanes BS-1 strain MTCC 25072)					
Banana	Eumusae leaf spot (<i>Mycosphaerella eumusae</i>)	-	5 litre/ha (Foliar spray)	750 Litre/ha	-
Major uses of fungicides					
		Dosage per ha			
Crop	Common name of the disease	a. i. (gm)	Formulation (g/ml)/%	Dilution in water (litre)	Waiting period from last application to harvest (days)
Copper oxychloride 50% WP					
Banana	Fruit rot	1.25	2.5	750-1000	-
	Leaf spot	1.25	2.5	750-1000	-
Mancozeb75%WP					
Banana	Cigar end rot, Tip rot, Sigatoka	1.125-1.5 kg	1.5-2kg	750 Ltr.	-
Pyraclostrobin 20% WG					
Banana	Sigatokaleaf spot	150	750	500	103
Metiram 55% + Pyraclostrobin 5% WG					
Banana	leaf spot	900-1050	1500-1750	500	85
Pyraclostrobin 133 g/l + Epoxiconazole 50g/l SE					
Banana	leaf spot	137.25	750	500	24
Major uses of insecticides					
Crop	Common Name of the pest	Dosage/ha		Dilution in water (litre)	Waiting period (days)
		a.i (gm)	Formulation (gm/ml)		
Carbofuran 03.00% CG					
Banana	Rhizome weevil	1g/ suckers	33g/sucker	-	-
	Aphid	50g/sucker	166g/sucker	-	-
	Nematode	1.5g/sucker	50 g/ sucker	-	-
Dimethoate 30.00% EC					
Banana	Aphid	0.03%	1485-1980	1500- 2000	At 3 months old
	Lace wing bug	0.03%	1485-1980	1500- 2000	-
Oxydemeton-methyl 25.00% EC					
Banana	Tingid bug	0.025%	1500-2000	1500- 2000	-
	Aphids	0.05%	3000-4000	1500- 2000	-
Quinalphos 25.00% EC					
Banana	Tingid bug	0.05%	3000-4000	500-1000	-

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